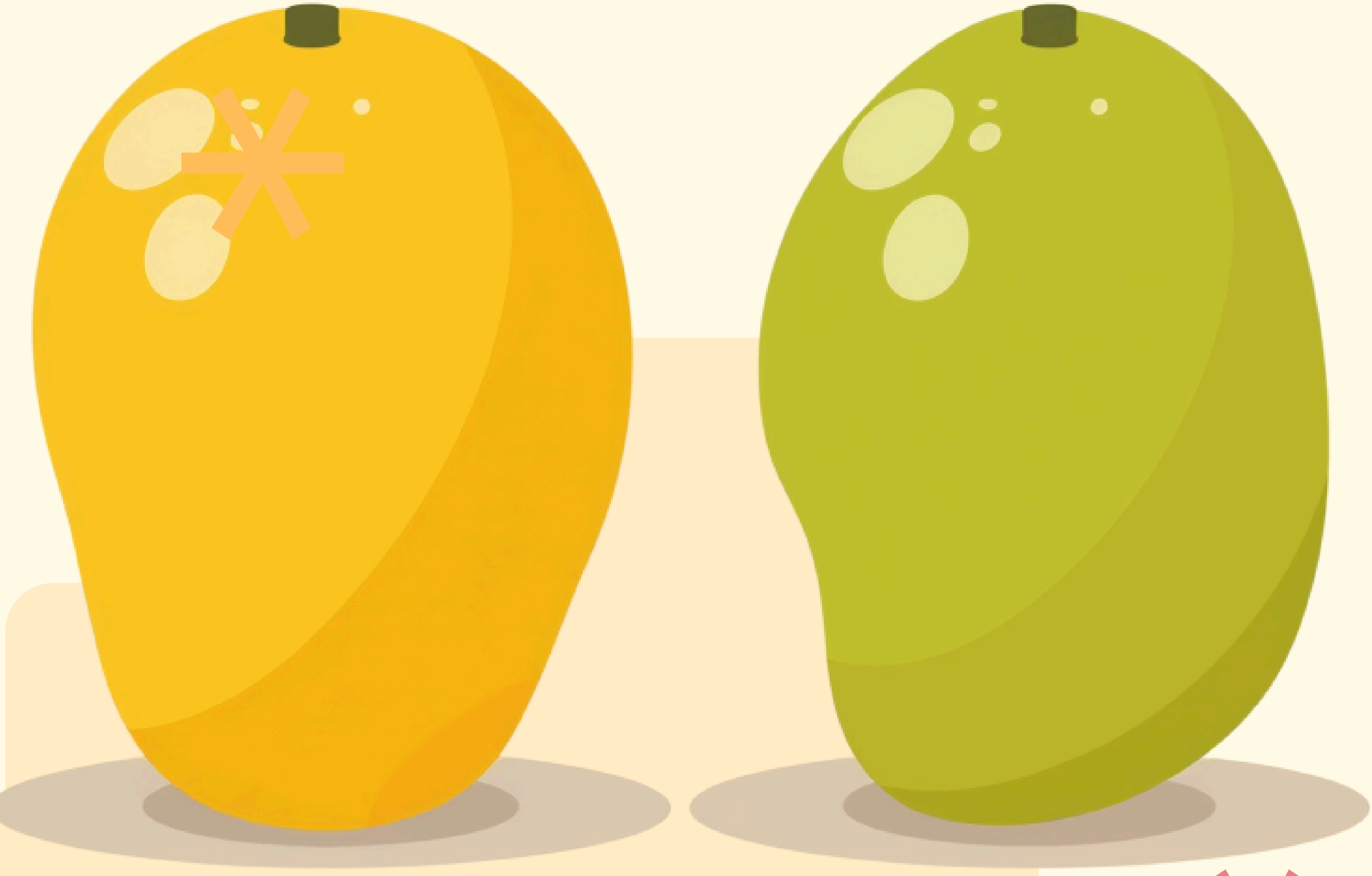
The Fruit Predictor: Your First Machine Learning Model

Build a simple ML model with scikit-learn

The Mystery Fruit

Can You Guess What This Fruit Is?

Here are your clues: Rough bumpy skin, many seeds inside, creamy sweet taste. Think like a detective! What fruit could this be?



Learning from Examples: Mango Varieties

How Computers Learn Like You Do

Machine Learning means learning from many examples instead of following rules. Imagine teaching someone to identify Alphonso vs. Langra mangoes by showing them 50 of each type. That's exactly how ML works!

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Features: The Clues to Identify Fruits

How Computers Learn to Recognize Objects

Feature 01

Weight – How heavy is the fruit? Heavier fruits often have more juice!

Feature 02

Texture – Is the skin smooth or rough? This helps tell fruits apart.

Feature 03

Color – What color do you see? Green, yellow, or red gives important clues.

Indian Example: At the Vegetable market, we choose a sweet watermelon by checking its weight and listening to the tapping sound. Computers use similar clues!

The 3-Step ML Cycle

Input (Features) → Model (Brain Builder) → Output (Prediction)

Follow these three simple steps to teach your computer to recognize fruits! Each step builds on the last to create a smart fruit predictor.

Step 1: Train

Show the computer many labeled fruit examples

Step 2: Test

Give it a new fruit to see if it guesses right

Step 3: Predict

Let it identify fruits by itself in real life!

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Meet scikit-learn: The Brain Builder

Your Toolbox for Teaching Computers

Think of scikit-learn as an inventor's toolbox! We feed clues (features) into the toolbox, and it creates a decision tree brain for us. Just like a chef uses ingredients to make delicious food!

What is a Decision Tree?

How Computers Make Fruit Decisions

A decision tree asks simple yes/no questions to identify fruits! Example: Is weight > 150g? If yes AND texture is smooth → Mango! If weight is light AND texture is smooth → Banana! It's like playing 20 questions with fruit clues.

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Step 01
Train with 5 fruit examples using weight, texture, and label

Step 02
Test with 1 new mystery fruit to check your model

Step 03
See if the model predicts the fruit correctly!

The Prediction Game: Train Your Model

Follow These Steps to Build Your Fruit Predictor

Ready to become a Machine Learning expert? Let's train your very own fruit prediction model step by step!

Training Example: 5 Fruits

Our Fruit Dataset for Teaching the Model

Here is our training data:

- Mango: 200g, Smooth, Yellow
- Banana: 120g, Smooth, Yellow
- Guava: 150g, Rough, Green
- Mango: 180g, Smooth, Orange
- Banana: 100g, Smooth, Yellow

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Predicting the 6th Fruit

The model uses what it learned to make an Anumaan (prediction)!

Before	**After**
New fruit is unknown to the model before prediction	Model predicts fruit name using learned clues from training

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Recap: What We Learned Today

Key Lessons from Our ML Journey

Today we discovered how computers can learn from examples, just like we do! The more examples we show, the smarter our model becomes at making predictions.

Point 01

Features are clues that help the model decide

Point 02

The 3-step process: Train, Test, Predict

Point 03

More examples make the model smarter!

Key Takeaways

Remember These 3 Important Points!

These are the main ideas from our Fruit Predictor journey. Keep them in mind as you explore more Machine Learning!

Takeaway 01

Features Help Computers Understand Objects

Takeaway 02

Training Teaches the Computer Patterns

Takeaway 03

Prediction Lets the Computer Guess New Things

Ready to Build Your Own Brain?

You're Now a Young ML Explorer!

Now you know how to teach a computer like a Vishleshak (analyst) and make Anumaan (predictions)! Keep exploring more fun ML projects and discover what else you can teach computers to do!

Thank You!

Keep asking questions and stay curious about Machine Learning!